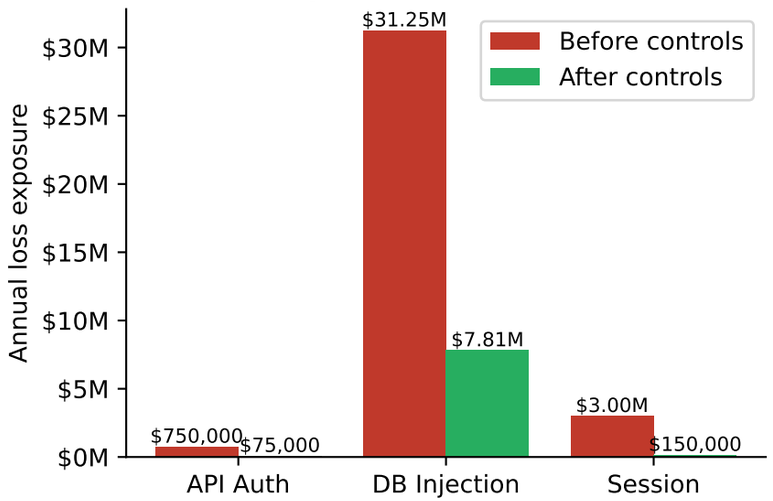
SecureBank Financial: Cyber Risk Quantification Dashboard

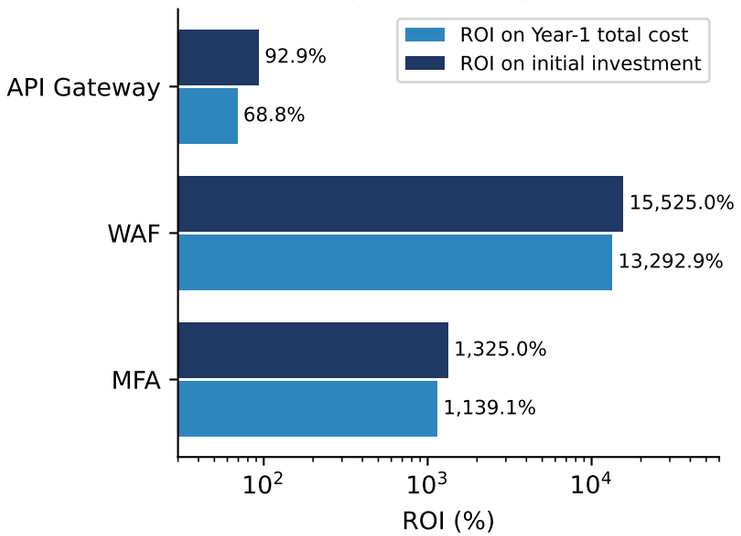
Chioma Ameh | C11/26/CGRCE/17567 | GRC103 Lab 3B | 29 Oct 2026 | Figures are annual, in US dollars

$35.00M	$8.04M	77.0%	$700,000	3,249%	9.5 days
Total annual exposure	Exposure after controls	Risk reduction	Investment (initial)	Year-1 ROI	Portfolio payback

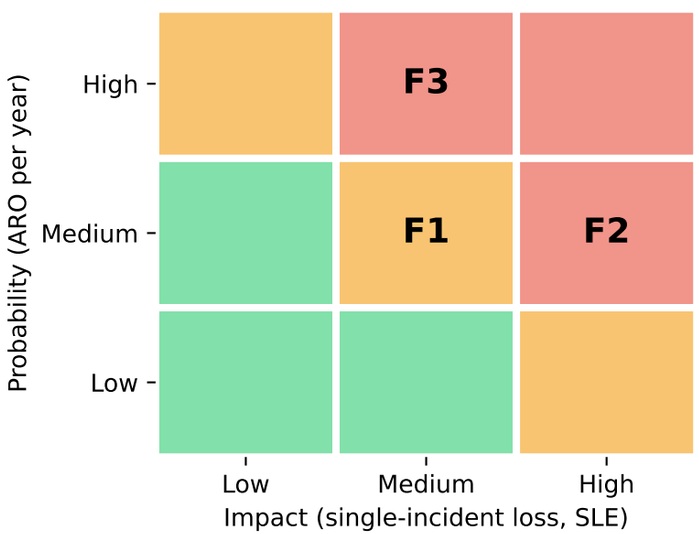
Exposure before vs after



ROI by control (log scale)



Risk heat map



Top 3 risks by annual loss

Risk	Annual loss	Priority
1. Database Injection Single incident $125.00M at 25% yearly probability	**$31.25M**	**High**
2. Session Hijacking Single incident $7.50M at 40% yearly probability	**$3.00M**	**High**
3. API Authentication Bypass Single incident $5.00M at 15% yearly probability	**$750,000**	**Medium**

Recommended controls

Control	Cost / yr maint.	Reduces	Payback
WAF Mitigates risk 1: Database Injection	$150,000 / $25,000	**$23.44M**	2.3 days
MFA Mitigates risk 2: Session Hijacking	$200,000 / $30,000	**$2.85M**	25.9 days
API Gateway Mitigates risk 3: API Authentication Bypass	$350,000 / $50,000	**$675,000**	6.7 months

Basis: ALE = SLE x ARO; ARO = exploit probability per year (assumed). Unverified inputs: loss figures and effectiveness are as supplied in the lab sheet. See workbook for assumptions.